

DATA ENGINEERING

MARVEL COMICS

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Preguntas Clave

- Universos y personajes
- Editorial y artistas
- Mercado y evolución temporal

No solo buscamos responder preguntas,
si no analizar todo el ecosistema.

League of Comic Geeks

Recopilar y analizar datos de 50,000 cómics y 30.000 personajes, para obtener artistas y su impacto histórico y financiero en el universo Marvel.

Web scrapping con Python:
Cloudscraper y BeautifulSoup



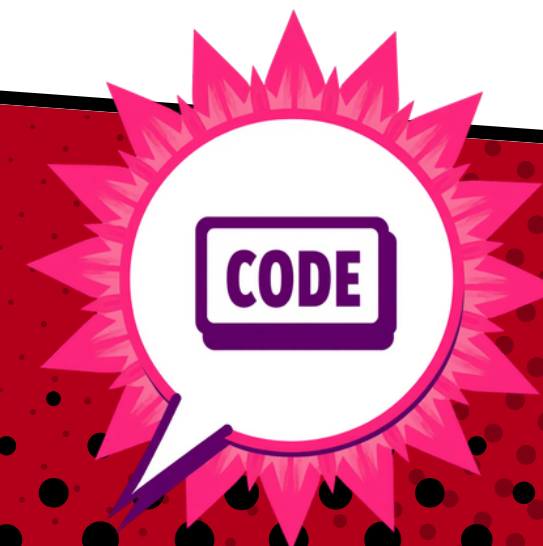
Arquitectura Técnica



Snowflake como
almacén central de
datos en la nube



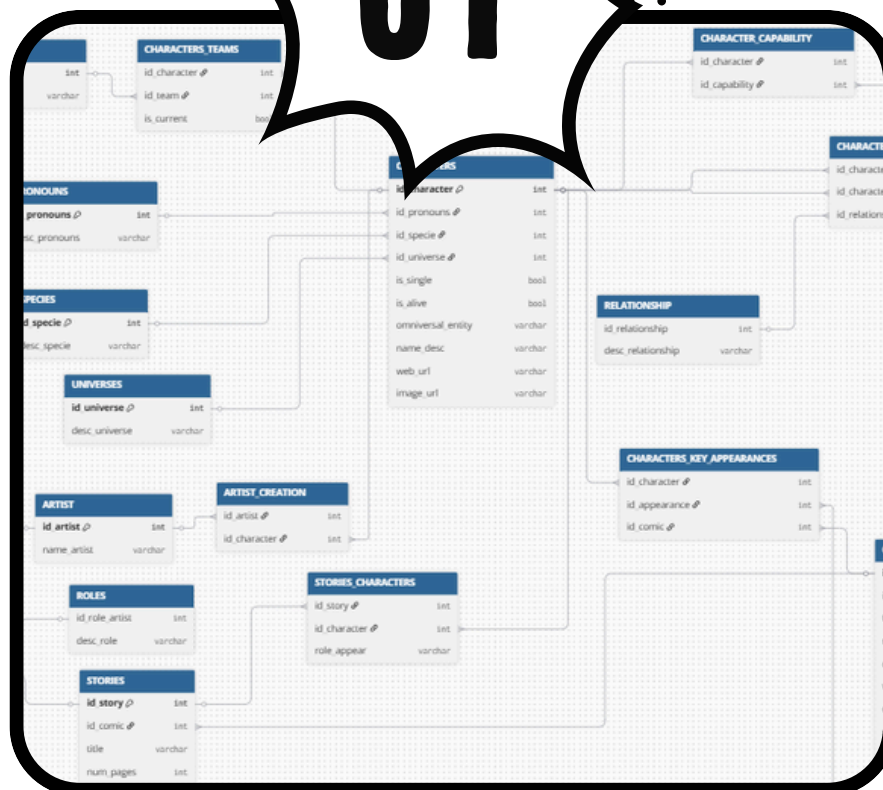
dbt para
transformación,
modelado de datos,
entornos y
documentación



Git para control de
versiones,
facilitando
despliegues (CI)

Ingesta de Datos

01



02

```

12 #!/usr/bin/env python3
13 INPUT_ISSUES_350M = "LOCAL_marvel_issue"
14 OUTPUT_FOLDER = "LOCAL_comics_scraped"
15 CI_SESSION = "123afcealedbe80cf7433a3"
16
17 SINGLE_URL = "https://leagueofcomicgeeks.com/comics/1234567890/"
18 SINGLE_URL = None # Cambia a None por la URL de la Comic que quieras
19 LIMIT = 0 # 0 para sin limit
20
21 if not os.path.exists(OUTPUT_FOLDER):
22     os.makedirs(OUTPUT_FOLDER)
23
24 def alert_window(titulo, mensaje, tipo="info"):
25     """Muestra un pñido y una ventana emergente de Windows."
26     # titulo: (Frecuencia, Duración)
27     if tipo == "error":
28         winsound.Beep(400, 200)
29     winsound.Beep(400, 300)
30     icon = 0x10 # Icono de Error (X roja)
31 else:
32     winsound.Beep(400, 200)
33     winsound.Beep(800, 200)
34     icon = 0x04 # Icono de Información (i azul)
35
36 ctypes.windll.user32.MessageBoxW(0, mensaje, titulo, icon)
37
38 def parse_comic_details(html):
39     soup = BeautifulSoup(html, "html.parser")
40
41     # --- 1. TÍTULO Y PORTADA ---
42     title_tag = soup.find("h1")
43     title = title_tag.get_text(strip=True) if title_tag else "unknown Title"
44     cover_tag = soup.find("div", class="cover-art")
45     cover_url = cover_tag.get_text(strip=True) if cover_tag and cover_tag.find("img") else ""
46
47     # --- 2. RATINGS Y STATS ---
48     rating_avg = "0"
49     avg_tag = soup.find("span", class="rating-avg")
50     if avg_tag: rating_avg = avg_tag.get_text(strip=True)
51
52     ratings_count = "0"
53     rc_container = soup.find("div", class="comic-score")
54     if rc_container:
55         count_match = re.search(r'(\d+)=ratings', rc_container.get_text())
56         if count_match: ratings_count = count_match.group(1)
57
58     stats = {"have": "0", "read": "0", "wish": "0"}
59     for container in soup.find_all("div", class="copy-medium mr-4"):
60         icon = container.find("img")
61         val_span = container.find("span", class="wl-1")
62         if icon and val_span:
63             classes = icon.get("class")
64             if "cg-icon-collect" in classes: stats["have"] = val
65             elif "cg-icon-readlist" in classes: stats["read"] = val
66             elif "cg-icon-wishlist" in classes: stats["wish"] = val
67
68     # --- 3. DETALLES TÉCNICOS ---
69     details = {"format": "Paper", "series": "0", "age": "0", "sku": "0", "cover_data": ""}
70     meta_tag = soup.find("div", class="re-compile(cg-copy-small, font-italic)")

```

03

```
windows PowerShell
X + v

indows PowerShell
opyright (C) Microsoft Corporation

install the latest PowerShell for

/aka.ms/PSWindows

PS A:\Xupi\Documents\Civica_curso\Web_Scrapper\SG
Procesando 10 elementos...
[1/10] Scrapeando: https://leagueofcomicgeeks.com/magazine/7976156/clive-baileys-hellraiser-summer-special-1
Guardado.
Esperando 0.78s...
[2/10] Scrapeando: https://leagueofcomicgeeks.com/comic/6499311/all-teen-comics-20
Guardado.
Esperando 0.73s...
[3/10] Scrapeando: https://leagueofcomicgeeks.com/comic/5413686/frankie-comics-11
Guardado.
Esperando 0.85s...
[4/10] Scrapeando: https://leagueofcomicgeeks.com/comic/1326395/frankie-comics-10
Guardado.
Esperando 0.73s...
[5/10] Scrapeando: https://leagueofcomicgeeks.com/comic/5141287/frankie-comics-9
Guardado.
Esperando 0.88s...
[6/10] Scrapeando: https://leagueofcomicgeeks.com/comic/9958844/frankie-comics-8
Guardado.
Esperando 0.68s...
[7/10] Scrapeando: https://leagueofcomicgeeks.com/comic/9081492/frankie-comics-7
Guardado.
Esperando 0.65s...
[8/10] Scrapeando: https://leagueofcomicgeeks.com/comic/7318925/frankie-comics-6
Guardado.
Esperando 0.77s...
[9/10] Scrapeando: https://leagueofcomicgeeks.com/comic/3198921/frankie-comics-5
Guardado.
Esperando 0.81s...
[10/10] Scrapeando: https://leagueofcomicgeeks.com/comic/7957934/frankie-comics-4
Guardado.
Esperando 0.85s...
```

04

The screenshot shows the Oracle SQL Developer interface. On the left, the 'Object Explorer' pane displays the database structure. The 'DEV_MARVEL_STAGE' table is selected under the 'MARVEL_STAGE' schema. The right pane shows the table's contents, which include columns 'NAME' and 'LOGC'. The table contains 83 rows of data, with the first few rows visible.

NAME	LOGC
LOGC_marvel_series.json	
LOGC_issue_9993509.json	
LOGC_issue_9994532.json	
LOGC_issue_9995970.json	
LOGC_issue_9997694.json	
LOGC_issue_9998486.json	
LOGC_issue_9998825.json	
LOGC_issue_9998751.json	
LOGC_issue_9995473.json	
LOGC_issue_9996892.json	

Problemas encontrados

01

Datos no estructurados (JSON):
→ Uso de LATERAL FLATTEN.

02

Datos muy poco fiables:

- Limpieza de datos en 21 modelos.
- Macros para estandarizar texto e ids.

03

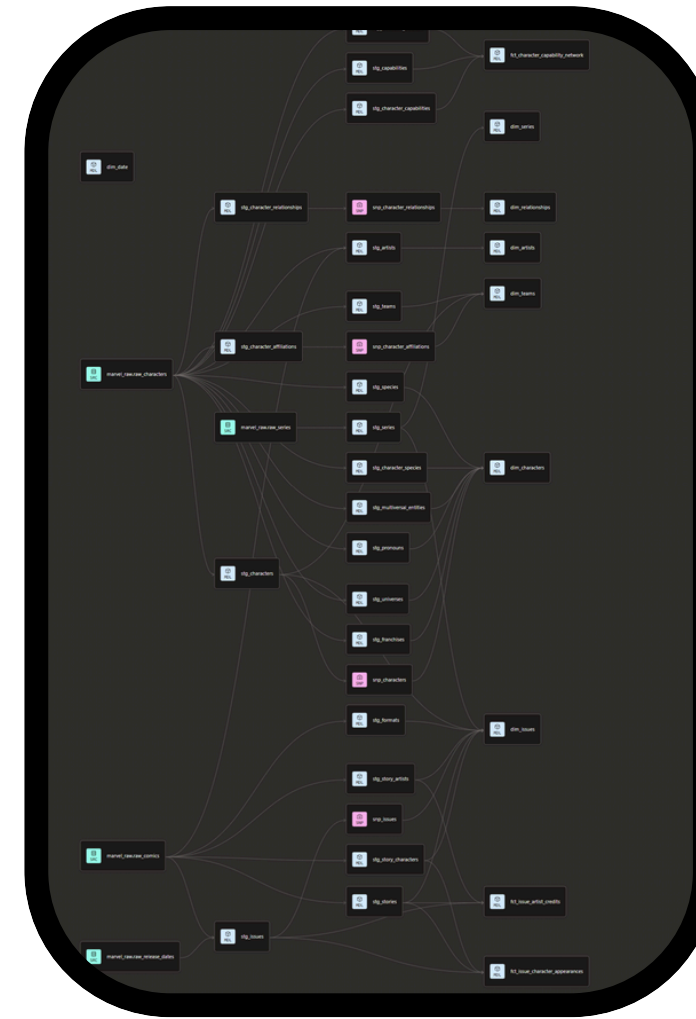
Duplicados con distinta marca de tiempo:
→ Uso de QUALIFY ROW_NUMBER().

04

Multiverso de Marvel pero no de otras franquicias:
→ Lógica de entidad multiversal común entre personajes.

05

Modelado difícil (todo en torno a entidad ISSUE):
→ Uso de snapshots y métricas en dimensiones.



Macros



marvel_data_engineering > macros > clean_text.sql

```
1  {# MACRO PARA ESTANDARIZAR FORMATO DE TEXTO #}  
2  
3  {% macro clean_text(texto) -%}  
4      upper(trim(nullif(cast({{ texto }} as string), '')))  
5  
6  {%- endmacro %}
```

marvel_data_engineering > macros > generate_marvel_id.sql

```
1  {# MACRO PARA GENERAR IDs ÚNICOS LLAMANDO A CLEAN_TEXT #}  
2  
3  {% macro generate_marvel_id(field_list) -%}  
4      {# 1. Aseguramos que sea una lista para poder iterar #}  
5      {%- set field_list = [field_list] if field_list is string else field_list -%}  
6  
7      {# 2. Creamos la lista de campos ya limpios usando la otra macro #}  
8      {%- set cleaned_fields = [] -%}  
9      {%- for field in field_list -%}  
10         {# Inyectamos el resultado de la macro clean_text #}  
11         {%- do cleaned_fields.append(clean_text(field)) -%}  
12     {%- endfor -%}  
13  
14     {# 3. Generamos el hash con dbt_utils #}  
15     {{ dbt_utils.generate_surrogate_key(cleaned_fields) }}  
16 {%- endmacro %}
```


Snapshot



```
marvel_data_engineering > snapshots > marts > core > snp_character_affiliations.sql

1  {% snapshot snp_character_affiliations %}
2
3  {{
4      config(
5          unique_key="character_id || '-' || team_id",
6          strategy='check',
7          check_cols=[
8              'character_id',
9              'team_id',
10             'is_current'
11         ],
12         hard_deletes='invalidate'
13     )
14 }}
15
16 select *
17 from {{ ref('stg_character_affiliations') }}
18
19 {% endsnapshot %}
```

Incremental



```
1  -- fct_issue_artist_credits.sql
2
3  {{
4      config(
5          materialized='incremental',
6          unique_key=['issue_id', 'artist_id', 'role_name'],
7          on_schema_change='fail',
8          incremental_strategy='merge'
9      )
10 }}
11
12  Preview CTE
13  with issues as (
14      select * from {{ ref('stg_issues') }}
15      {% if is_incremental() %}
16          where loaded_at > (select coalesce(max(loaded_at), '1900-01-01') from {{ this }})
17      {% endif %}
18  ),
```

Demostración



[https://github.com/Axgaldo/
marvel_data_engineering](https://github.com/Axgaldo/marvel_data_engineering)